

=====

PART A : Final Project Proposal

=====

Project Overview

This project is a decentralized crowdfunding platform built on Solana that releases funds to developers only after milestone completion verified through community voting. Instead of trusting developers upfront, all contributions are held in escrow and paid out in small tranches as milestones are approved. The platform uses weighted and bicameral voting to balance financial stake with community consensus, reducing the risk of fraud and low-quality delivery. By structurally bounding risk and decentralizing verification, the platform aims to create a safer and more accountable alternative to traditional crowdfunding.

Core Value Proposition & Product-Market Fit (PMF)

The core value proposition is **risk-bounded, milestone-based crowdfunding** that aligns developer incentives with funder protection without relying on centralized intermediaries. Unlike traditional platforms where funds are released upfront, this system escrows all contributions and releases them only when predefined milestones are verified through community voting. This reduces **rug-pull risk, increases accountability, and creates a transparent feedback** loop between developers and funders. The product fits markets where trust, capital efficiency, and delivery accountability are critical—such as open-source development, indie software projects, and early-stage Web3 teams—by offering a governance-native alternative to Kickstarter-style crowdfunding.

Key value areas:

1. *Escrowed milestone funding instead of upfront payout*
 2. *Decentralized verification through voting*
 3. *Bounded trust model that limits worst-case funder loss*
-

Key Target Markets

1. *Open-source developers seeking non-VC funding with accountability*
 2. *Indie software founders building early-stage SaaS or tools*
 3. *Web3 builders launching decentralized applications*
 4. *Crypto-native investors interested in governance-driven funding*
 5. *DAOs and grant programs seeking milestone-based capital allocation*
-

Competitor Landscape

AI-Identified Competitors

- **Kickstarter** – Traditional crowdfunding without milestone escrow
- **Gitcoin Grants** – Quadratic funding for open-source projects
- **Juicebox** – DAO treasury crowdfunding platform
- **Mirror.xyz** – Tokenized crowdfunding and publishing

Manually Researched Competitors

- **Angor** – Bitcoin-based decentralized crowdfunding protocol using time-locked staged funding and Nostr for metadata
 - **Giveth** – Ethereum-based donation and crowdfunding platform
-

Gap Analysis

Angor is the closest architectural analogue to this project. It uses Bitcoin timelocks and multisig scripts to enable staged funding and allows investors to recover unspent funds. However, it lacks native on-chain governance and milestone verification logic; fund release is driven by time progression rather than proof of work completion.

Most other platforms either:

- *Release funds upfront (Kickstarter, Mirror)*
- *Rely on centralized review committees (many grant programs)*
- *Focus on donation-style funding rather than escrow (Gitcoin, Giveth)*
- *Lack programmable milestone enforcement (Juicebox)*

No existing platform combines:

- *On-chain escrow*
- *Milestone-based releases*
- *Governance-based verification*
- *Bicameral or weighted voting*
- *Bounded trust for first-time developers*

This project differentiates itself by embedding verification and accountability directly into the funding flow, rather than treating them as off-chain or social processes.

Founder-Market Fit (FMF)

- I am a computer science graduate student with hands-on experience building Solana smart contracts, working with Anchor, and designing decentralized governance logic.

- My academic focus on distributed systems, cryptography, and blockchain protocols aligns directly with the technical challenges of escrow, voting, and milestone automation.
 - I am personally motivated by the problem of trust in decentralized funding, having explored governance, encryption, and Solana-based protocol design extensively.
 - This combination of technical depth, domain curiosity, and practical experimentation gives me a strong founder-market fit to prototype and evolve a governance-native crowdfunding platform.
-

=====

PART B: PROCESS APPENDIX

=====

This section documents the full reasoning process, AI interactions, independent research, adversarial critique, and design refinements.

Step 1: Initial Idea

Initial	Concept	Summary
A Solana-based crowdfunding platform where developers raise funds for projects, but money is stored in a vault and released in chunks only after milestone completion verified by funder voting. The system initially used weighted voting and retry limits for milestone approval.		

Reflection

The core intuition was that most crowdfunding failures happen because funds are given too early. By turning funding into a governance-driven escrow process, the platform reduces trust requirements and bounds downside risk.

Step 2: AI Prompt – Value Proposition & PMF

Prompt Used

“Based on my idea of a milestone-based crowdfunding platform built on Solana—where funds are escrowed and released only after milestone completion verified through community voting—help outline the core value proposition and initial thoughts on product–market fit. What are two or three key value areas this platform provides?”

AI Output (Summarized)

- Emphasized accountability and trust minimization
- Highlighted open-source and Web3 as primary markets

- Identified escrow + milestone releases as differentiators

Manual

Synthesis

Refined the value proposition into a single framing:

“Risk-bounded crowdfunding through milestone-based escrow and decentralized verification.”

Step 3: AI Prompt – Target Markets

Prompt Used

“For this value proposition—milestone-based escrow crowdfunding with decentralized governance—suggest two to five key target demographics or market segments that are most likely to benefit from this platform.”

AI Output (Summarized)

- Open-source developers
- Web3 startups
- Indie SaaS founders
- DAOs and grant programs

Manual

Refinement

Added “crypto-native investors” as a distinct segment to capture governance-oriented funders.

Step 4: AI Prompt – Competitor Analysis

Prompt Used

“Identify key competitors for a project with this value proposition targeting these markets. What are potential weaknesses in their offerings, and where might this platform differentiate itself?”

AI Output (Summarized)

- Kickstarter: no escrow or milestone enforcement
- Gitcoin: no milestone verification logic
- Juicebox: governance complexity and no staged escrow
- Mirror: upfront token sales

Step 5: Manual Research (Mandatory)

Manual	Findings
Researched the following platforms independently:	
● Angor	
● Gitcoin	
● Giveth	
● Juicebox	

Key	Discovery
Angor emerged as the closest real-world analogue. It enables staged crowdfunding on Bitcoin using timelocks and multisig, and uses Nostr for decentralized metadata. However, it lacks milestone verification via governance and relies on time progression instead of proof of work completion.	

Gap	vs	AI	Findings
The AI did not identify Angor, which turned out to be the most relevant competitor. This demonstrated the importance of independent research and materially refined the competitive analysis.			

Step 6: Founder-Market Fit Draft

Initial FMF Focus		
● Solana	program	development
● Anchor	framework	experience
● Governance	logic	design

Step 7: Adversarial AI Critique

“Critique my project’s value proposition, target markets, and competitive analysis (pasted below). Why might this not represent a true blue-ocean opportunity? What types of competitors or adjacent solutions might have been overlooked?”

AI Critique (Summarized)

- Crowdfunding is a saturated category
- Web3 adoption is still niche
- UX complexity may deter mainstream users
- Trust improvements alone may not drive adoption
- Grant platforms and accelerators were under-considered

My Analysis of Validity

These critiques were valid. In particular, the discovery of Angor confirmed that staged crowdfunding already exists in decentralized form. This pushed the project toward clearer differentiation through governance-based milestone verification and bounded trust design, rather than claiming full novelty.

Step 8: Cold Start Analysis & Design Decision

Problem

Identified

How can funders trust new developers without third-party identity, staking requirements, or centralized moderators?

Initial Over-Engineered Direction

- *Tier-based reputation systems*
- *Developer staking requirements*
- *Advancement*

heuristics

Final

Simplified

Decision

Rather than adding heavy governance complexity:

- *Trust new developers for only one milestone*
- *Bound loss via escrow*
- *Cap first milestone payouts*
- *Cap total raise size for first-time developers*

Rationale

This preserves security while keeping the protocol simple, auditable, and shippable. It mirrors Angor's "bounded trust" philosophy while adding richer verification semantics through voting.

Step 9: Bicameral + Weighted Voting Refinement

Final Voting Model

- Chamber A: Quadratic-weighted voting (voting power = $\sqrt{\text{funded amount}}$)
- Chamber B: Headcount voting (1 wallet = 1 vote)
- A milestone passes only if both chambers approve

Why

This

Matters

This prevents:

- *Whale domination*
- *Mob veto attacks*
- *Rubber-stamping*

of

poor

work

Step 10: Final Refinement Summary

Changes Made

- Integrated Angor into competitor analysis
- Removed staking and tier systems
- Simplified cold-start handling
- Focused on bounded trust
- Strengthened PMF clarity
- Tightened market scope to Web3 + open-source builders

Why

These

Changes

Were

Made

To keep the system:

- Minimal
- Secure
- Explainable
- Aligned with MVP constraints
- Clearly differentiated from both Web2 crowdfunding and Bitcoin-native approaches like Angor